Project Title

MediSearch

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Medi Search is an intelligent patient record search and retrieval system designed to help doctors efficiently find relevant medical information from large collections of patient records. The system applies advanced Data Structures and Algorithms to perform efficient medical record searching, fuzzy matching, and similar-case identification. KMP, Rabin-Karp, and Aho-Corasick algorithms are used for exact and multiple-pattern searches across patient records. Levenshtein Distance with Dynamic Programming helps identify search terms containing spelling errors. For similar patient cases, TF-IDF and Cosine Similarity are used to compare medical histories and rank records based on their similarity. The project demonstrates how different algorithmic techniques can be selected according to different search requirements, improving retrieval efficiency and scalability.

Signature of the Guide